1 **(a)** $y = kx^2$ or $36 = k(3^2)$ **[M1]**

$4x^2$ **[A1]**

(b) 100 **[B1]**

[Total Marks: 3]

2 $y = k(x + 2)$ **[M1]**

$20 = k(3 + 2)$ leading to $k = 4$ **[M1]**

36 **[A1]**

[Total Marks: 3]

3 **(a)** M1 for $y = kx^2$ or $36 = k(3)^2$

$4x^2$ **[A1]**

(b) M1 for $64 = 4x^2$ leading to $x^2 = 16$

4 and -4 (or ± 4) **[A1]**

[Total Marks: 4]

4 **(a)** Sets up equation $y = kx$ and substitutes $x = 5, y = 15$ to get $15 = 5k$ **[M1]**

$y = 3x$ **[A1]**

(b) Substitutes $x = 12$ into their equation from part (a), e.g. $y = 3 \times 12$ **[M1]**

36 **[A1]**

[Total Marks: 4]

5 **(a)** Substitutes into the proportion equation to find the constant, e.g. $135 = k(1 + 2)^3$ **[M1]**

$5(x + 2)^3$ **[A1]**

(b) 40 **[B1]**

(c) Sets up the equation $320 = 5(x + 2)^3$ and simplifies to $(x + 2)^3 = 64$ **[M1]**

2 **[A1]**

[Total Marks: 5]

6 **(a)** $y = kx^2$ leading to $36 = k(3)^2$ **[M1]**

$4x^2$ **[A1]**

(b) 100 **[B1]**

(c) $196 = 4x^2$ leading to $x^2 = 49$ **[M1]**

7 and -7 **[A1]**

[Total Marks: 5]

7 **(a)** $y = kx^3$ **[M1]**

$24 = k(2)^3 \implies k = 3$ **[M1]**

$3x^3$ **[A1]**

(b) $y = 3(5)^3 = 3(125)$ **[M1]**

375 **[A1]**

(c) $81 = 3x^3$ **[M1]**

$x^3 = 27$ **[M1]**

3 **[A1]**

[Total Marks: 5]

8 **(a)** Recognises inverse square proportion, e.g. tests p^2q **[M1]**

$2^2 \times 100 = 400$, $4^2 \times 25 = 400 \implies k = 400$ **[M1]**

$\dfrac{400}{p^2}$ **[A1]**

(b) $q = \dfrac{400}{5^2} = \dfrac{400}{25}$ **[M1]**

16 **[A1]**

(c) $1 = \dfrac{400}{p^2}$ **[M1]**

$p^2 = 400$ **[M1]**

20 (Accept ± 20) **[A1]**

[Total Marks: 5]

9 **(a)** Sets up $P = k_1\sqrt{Q}$ and substitutes $Q = 9, P = 12 \implies 12 = k_1\sqrt{9} \implies 12 = 3k_1 \implies k_1 = 4$, so $P = 4\sqrt{Q}$ **[M1]**

Sets up $Q = \dfrac{k_2}{R^3}$ and substitutes $R = 2, Q = 2 \implies 2 = \dfrac{k_2}{2^3} \implies 2 = \dfrac{k_2}{8} \implies k_2 = 16$, so $Q = \dfrac{16}{R^3}$ **[M1]**

Substitutes Q into the P equation: $P = 4\sqrt{\dfrac{16}{R^3}}$ **[M1]**

Simplifies the square root: $P = 4 \times \dfrac{4}{\sqrt{R^3}}$ **[M1]**

$P = \dfrac{16}{R^{1.5}}$ or $P = 16R^{-1.5}$ or $P = \dfrac{16}{R\sqrt{R}}$ **[A1]**

(b) Substitutes $R = 4$ into their formula: $P = \dfrac{16}{4\sqrt{4}}$ or $P = \dfrac{16}{8}$ **[M1]**

2 **[A1]**

[Total Marks: 7]

10 **(a)** M1 for setting up $A = kB^2$ and finding $k = 5$ (i.e., $A = 5B^2$)

M1 for setting up $B = \dfrac{m}{C+2}$ and finding $m = 12$ (i.e., $B = \dfrac{12}{C+2}$)

M1 for substituting $C = 4$ into their equation for B to get $B = 2$

20 **[A1]**

(b) M1 for substituting their expression for B into their equation for A: $A = 5\left(\dfrac{12}{C+2}\right)^2$

M1 for expanding or rearranging to isolate $(C+2)^2$: $(C+2)^2 = \dfrac{720}{A}$

M1 for taking the positive square root: $C + 2 = \sqrt{\dfrac{720}{A}}$

$\sqrt{\dfrac{720}{A}} - 2$ (or equivalent valid form e.g. $\dfrac{12\sqrt{5}}{\sqrt{A}} - 2$) **[A1]**

[Total Marks: 8]

11 **(a)** Sets up equation $y = \dfrac{k}{x^2}$ **[M1]**

Substitutes $x = 4, y = 3$ to get $3 = \dfrac{k}{4^2} \implies k = 3 \times 16 = 48$ **[M1]**

$y = \dfrac{48}{x^2}$ **[A1]**

(b) Sets $0.75 = \dfrac{48}{x^2}$ **[M1]**

Rearranges to $x^2 = \dfrac{48}{0.75} = 64$ **[M1]**

$x = 8, x = -8$ (both required for final mark) **[A1]**

[Total Marks: 6]

12 **(a)** $P = \dfrac{k}{Q^2}$ **[M1]**

$1.5 = \dfrac{k}{4^2}$ **[M1]**

$\dfrac{24}{Q^2}$ **[A1]**

(b) 0.24 or $\dfrac{6}{25}$ **[B1]**

(c) $24 = \dfrac{24}{Q^2}$ or $Q^2 = 1$ **[M1]**

1 **[A1]**

[Total Marks: 6]

13 **(a)** **(i)** $P = \dfrac{k}{\sqrt{Q}}$ leading to $12 = \dfrac{k}{\sqrt{9}}$ **[M1]**

$k = 36$ **[B1]**

3 **[A1]**

(ii) $1.5 = \dfrac{36}{\sqrt{Q}}$ leading to $\sqrt{Q} = 24$ **[M1]**

576 **[A1]**

(b) Identifying that the new value of P involves dividing by $\sqrt{4}$ or 2 **[M1]**

50 **[A1]**

[Total Marks: 7]

14 **(a)** $M = \dfrac{k}{\sqrt{N}}$ or $15 = \dfrac{k}{\sqrt{16}}$ **[M1]**

$\dfrac{60}{\sqrt{N}}$ **[A1]**

(b) $\dfrac{60}{\sqrt{100}}$ or $\dfrac{60}{10}$ **[M1]**

6 **[A1]**

(c) $1.2 = \dfrac{60}{\sqrt{N}}$ **[M1]**

$\sqrt{N} = \dfrac{60}{1.2}$ or $\sqrt{N} = 50$ **[M1]**

2500 **[A1]**

[Total Marks: 7]

15 **(a)** M1 for recognising that y is inversely proportional to x^2 (e.g. testing $x^2y = c$)

M1 for a correct substitution to find k, e.g., $k = 4^2 \times 9 = 144$

$\dfrac{144}{x^2}$ **[A1]**

(b) M1 for $2.25 = \dfrac{144}{x^2}$ or $x^2 = 64$

8 **[A1]**

(c) M1 for substituting $3x$ into their equation: $y_{new} = \dfrac{144}{(3x)^2} = \dfrac{144}{9x^2}$

$\dfrac{1}{9}$ (or 0.111...) **[A1]**

[Total Marks: 7]

16 **(a)** Sets up the correct proportional relationship, e.g. $p = \dfrac{k}{\sqrt{q}}$ or $5 = \dfrac{k}{\sqrt{144}}$ **[M1]**

$p = \dfrac{60}{\sqrt{q}}$ **[A1]**

(b) Substitutes $p = 15$ into their equation, e.g. $15 = \dfrac{60}{\sqrt{q}}$ or $\sqrt{q} = 4$ **[M1]**

16 **[A1]**

(c) Recognises that increasing by 300% means the new value of q is $4q$ **[B1]**

Substitutes $4q$ into the equation to find the new p: $p_{new} = \dfrac{60}{\sqrt{4q}} = \dfrac{60}{2\sqrt{q}} = \dfrac{1}{2}p$ **[M1]**

50 **[A1]**

[Total Marks: 7]

17 (a) Substitutes into the proportion equation to find the constant, e.g. $4 = \dfrac{k}{\sqrt{36}}$ **[M1]**

$\dfrac{24}{\sqrt{Q}}$ **[A1]**

(b) 2 **[B1]**

(c) Sets up the equation $1.5 = \dfrac{24}{\sqrt{Q}}$ leading to $\sqrt{Q} = 16$ **[M1]**

256 **[A1]**

(d) Identifies the new value of Q is $4Q$, or uses numerical values (e.g. Q increases from 36 to 144) **[M1]**

Evaluates the new value of P as $\dfrac{1}{2}P$, or calculates the new P is 2 when original was 4 **[M1]**

50 **[A1]**

[Total Marks: 8]

18 (a) Tests products or ratios, e.g., shows $x^2y = 300$ for all pairs or identifies $k = 300$ **[M1]**

Identifies $y \propto \dfrac{1}{x^2}$ as the correct model **[A1]**

$\dfrac{300}{x^2}$ **[A1]**

(b) Uses 1.5 multiplier and finds $(1.5)^3$ or 3.375 **[M1]**

Identifies increase factor as $3.375 - 1$ or 2.375 **[M1]**

237.5 **[A1]**

[Total Marks: 6]

19 (a) Uses new $x = 1.2x$ to write $y_{new} = k(1.2x)^3$ **[M1]**

Calculates $1.2^3 = 1.728$ **[M1]**

72.8 **[A1]**

(b) Tests one incorrect model and shows the constant is not equal, e.g. for $A \propto v$, calculates

$\frac{12}{2} = 6$ but $\frac{48}{4} = 12$ **[M1]**

Tests $A \propto v^2$ and shows $\frac{A}{v^2}$ is constant by calculating $\frac{12}{4} = 3$, $\frac{48}{16} = 3$, and $\frac{300}{100} = 3$ **[M1]**

$A \propto v^2$ **[A1]**

[Total Marks: 6]

20 (a) (i) Recognises that x^2y is constant (128), or identifies the model as $y \propto \frac{1}{x^2}$ **[M1]**

Sets up the equation $y = \frac{k}{x^2}$ and correctly calculates $k = 128$ **[M1]**

$\frac{128}{x^2}$ **[A1]**

(ii) 1.28 **[B1]**

(b) Sets up the correct scale factor equation for B, e.g. $m^3 = 1.728$ or $\left(1 + \frac{x}{100}\right)^3 = 1.728$ **[M1]**

Calculates the multiplier $m = \sqrt[3]{1.728} = 1.2$ **[M1]**

20 **[A1]**

[Total Marks: 7]

21 (a) Forms equation: $y = \frac{k}{\sqrt{x}}$ **[M1]**

Uses multiplier 1.44 for the increase in x: New $y = \frac{k}{\sqrt{1.44x}}$ **[M1]**

Simplifies denominator: $\sqrt{1.44} = 1.2$, so New $y = \frac{1}{1.2} \times \frac{k}{\sqrt{x}}$ (or $\frac{5}{6}y$) **[M1]**

Calculates percentage change: $\left(\frac{5}{6} - 1\right) \times 100$ **[M1]**

-16.7 (or 16.7 decrease) **[A1]**

(b) Deduces relationship between P and R: $P = k_1\left(\frac{k_2}{R}\right)^3 = \frac{K}{R^3}$ (or $P \propto \frac{1}{R^3}$) **[M1]**

Substitutes given values to find constant: $10 = \frac{K}{2^3} \implies K = 80$ **[M1]**

Substitutes $R = 4$ into their equation: $P = \frac{80}{4^3} = \frac{80}{64}$ **[M1]**

1.25 (or $\frac{5}{4}$) **[A1]**

[Total Marks: 7]

22 (a) Tests a non-linear variation model and sets up an equation, e.g. $y = \frac{k}{x^2}$ **[M1]**

Correctly identifies the constant k by substituting a coordinate pair, e.g.

$100 = \frac{k}{2^2} \implies k = 400$ **[M1]**

$y = \frac{400}{x^2}$ **[A1]**

(b) (i) Sets up $z = kx$ and finds k: $10 = 4k \implies k = 2.5$ **[M1]**

$\frac{z}{2.5}$ or $\frac{2z}{5}$ or $0.4z$ **[A1]**

(ii) Substitutes their expression for x into the equation from part (a): $y = \frac{400}{(\frac{2z}{5})^2}$ or $y = \frac{400}{(0.4z)^2}$ **[M1]**

Squares the fraction or decimal correctly: $y = \frac{400}{\frac{4z^2}{25}}$ or $y = \frac{400}{0.16z^2}$ **[M1]**

$\frac{2500}{z^2}$ **[A1]**

[Total Marks: 8]

23 (a) Demonstrating that $y \times x^3$ is constant for at least two pairs (e.g., $40 \times 2^3 = 320$ and $5 \times 4^3 = 320$) **[M1]**

$y \propto \frac{1}{x^3}$ **[B1]**

$\frac{320}{x^3}$ **[A1]**

(b) $2560 = \frac{320}{x^3}$ leading to $x^3 = \frac{1}{8}$ **[M1]**

$\frac{1}{2}$ or 0.5 **[A1]**

(c) $x = \frac{k_2}{w}$ leading to $2 = \frac{k_2}{5}$ **[M1]**

$x = \frac{10}{w}$ **[A1]**

Substituting into their equation for y: $y = \frac{320}{(\frac{10}{w})^3}$ **[M1]**

$0.32w^3$ or $\frac{8}{25}w^3$ **[A1]**

[Total Marks: 9]